VaidyaVaani — Problem Statement & Solution

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Problem statement

VaidyaVaani is a voice interpreter for the consultation room: any patient, any doctor, each heard in their own language, with every dose read back safely.

Paste-ready version for the submission:

> In Bengaluru's government hospitals and urban health centres, patients and staff often share no language. A study published in July 2026 at a Bengaluru teaching hospital and two urban health centres found that every patient interviewed had trouble communicating with staff, 48% believed they were misdiagnosed or never diagnosed because of it, and 98% of participants said no formal interpreter was available. Today the interpreter is a relative, another patient or a security guard, and the prescription goes home in a language the patient cannot read.
>
> VaidyaVaani is a real-time, two-way voice interpreter built on SLNG. The patient speaks Kannada, Tamil, Telugu, Hindi, Bengali or Urdu; the doctor hears or reads it in English, and the reply comes back in the patient's own voice language. Medicines, doses and timings are never paraphrased: they are extracted, confirmed by the doctor, and read back to the patient, who then repeats them in their own words (teach-back) before they leave. Audio stays in India and nothing is stored without consent.

The one-line pitch: No patient in Bengaluru should need to bring their son to understand their own diagnosis.

The problem, with evidence

Language mismatch is the norm in Bengaluru clinics, not the exception, and it is causing missed diagnoses and wrongly taken medicine.

Bengaluru is India's most multilingual district

An analysis of the 2011 Census found 107 languages spoken in Bengaluru, the most of any district in India. Mother tongues: Kannada 44.5%, Tamil 15%, Telugu 14%, Urdu 12%, Hindi 6%, Malayalam 3% (Udayavani). Add the migrant workforce and staff recruited from other states, and the doctor and patient in a room often share nothing.

The Bengaluru study (July 2026) is our anchor

Kozan, Siddaiah and Dasgupta, *Frontiers in Communication* interviewed 25 patients and 25 health workers at a tertiary teaching hospital and two urban primary health centres in Bengaluru, speaking 11 languages.

Finding	Patients	Health workers
Had trouble communicating because of language	100%	96%
Believe language caused misdiagnosis or no diagnosis	48%	24% admit doing it
Quality of care affected	72%	–
Used informal interpreters (family, staff, other patients, a guard)	88%	68%
No health information available in their language	76%	–
Said no formal interpreter service existed	98% of all participants	

Three details from the study shape our design:

- **It is many-to-many, not Kannada vs Hindi.** Patients' mother tongues were mostly Tamil (48%) and Bengali (20%); health workers were mostly Malayalam speakers (40%). The tool must work for any pair.
- **Patients can't fall back on reading.** 56% of patients had not completed Class 10, so translated text alone fails. It has to be voice.
- **Existing apps were tried and abandoned.** 28% of health workers had used translation apps, but said Google Translate output often made no sense to patients, came out transliterated or overly formal, so they stopped.

What goes wrong in real consultations (from the study's interviews)

- A patient took tablets at the wrong times until her daughter read the prescription for her.
- A doctor mistook severe period cramps for acidity until a translator clarified.
- A patient with neck pain changed doctors four or five times to find one who spoke Hindi, getting worse meanwhile.
- Staff could not explain consent forms, costs, or how to use an injection or IUD. Patients reported paying for tests they didn't need when they went without their English-speaking son.
- Some patients skip care entirely if no one speaks their language.

It is also a live political flashpoint in Karnataka

- The Karnataka Language Development Authority proposed that government hospital prescriptions be written only in Kannada; doctors replied that many patients, including migrants, don't read Kannada and many doctors can't write it (South First, Sep 2024).
- A resident doctor at Minto Ophthalmic Hospital was confronted by activists for not speaking Kannada, and resident doctors protested, disrupting outpatient services at Victoria and Vanivilas hospitals (The News Minute, Deccan Herald).

VaidyaVaani takes no side in that fight: Kannada is a first-class language in the product, and so is every other patient's language.

Who it's for

The patient and the doctor use it together on one device; the hospital buys it.

User	Their moment of pain	What VaidyaVaani gives them
Patient (often older, low literacy, migrant or rural)	Can't describe symptoms, can't read the prescription, needs a relative to come	Speaks freely in their language; hears the plan and doses back in it; leaves with a voice note they can replay

User	Their moment of pain	What VaidyaVaani gives them
Doctor (often from another state, 2–4 minutes per patient in a busy OPD)	Guessing from gestures, hunting for a staff member to translate	English transcript of the patient on screen, their reply spoken in the patient's language, a prescription confirmed before it is read out
Nurse / pharmacist	Explaining injections, consent forms, dosing times, costs	The same interpreter at the counter, with ready phrase sets for common instructions
Family member	Has to take time off work to translate	Doesn't have to come
Hospital admin / Health Department	No interpreter budget; complaints and language disputes	A per-minute tool instead of hiring interpreters for 100+ languages; a log of which languages are needed

Why existing options fall short

Every current option either isn't there when needed, can't be trusted with a dose, or needs the patient to read.

Option	What it does well	Why it fails in a Bengaluru OPD
Family / staff / other patients as interpreters	Free, available sometimes	Untrained, miss medical terms, add their own interpretation, burden relatives, break privacy on sensitive topics
Professional interpreters	Accurate	Essentially none exist in Indian public hospitals (98% in the study said none available); can't cover 107 languages

Option	What it does well	Why it fails in a Bengaluru OPD
Google Translate on the doctor's phone	Free, many languages	Text-first, one direction at a time, no medical safeguards; staff in the study abandoned it because output confused patients. A JAMA study found 8% of Chinese discharge sentences translated by it had potential for significant harm, with errors tied to medical jargon (UCSF/JAMA via Becker's)
Bhashini (government language AI)	22 scheduled languages, now being plugged into Ayush and Ayushman Bharat platforms (Digital Health News)	A building block for portals and apps, not a ready consultation-room product with dose safety, teach-back and turn-taking. Complementary, not a competitor
Generic AI transcription	Fast notes	Can invent words: an AP investigation found Whisper inventing a non-existent medicine, "hyperactivated antibiotics" (AP via ABC News)

Our difference in one line: voice-first for low-literacy patients, any-to-any Indian languages, and a safety layer that treats drug names, doses and timings as data to confirm, not text to translate.

The solution

One phone or tablet on the doctor's desk, two big talk buttons, and three flows: interpret, prescribe safely, confirm understanding.

How a consultation runs

1. **Start.** The patient taps their side and says anything. The agent detects the language and confirms it out loud in that language ("You're speaking Tamil, correct?"). It also asks consent to use the interpreter, in that language.

2. **Interpret.** Patient speaks → doctor sees the English transcript and hears it if they want. Doctor speaks English or Hindi → patient hears it in their language. Push-to-talk per side, so the system always knows who is speaking.
3. **Red-flag watch.** If the patient mentions a danger sign (chest pain, breathlessness, bleeding in pregnancy, thoughts of self-harm), the doctor's screen flags it at once, even mid-sentence.
4. **Prescribe safely.** The doctor dictates the prescription. The agent pulls out each medicine, dose, timing, food relation and duration into a card on screen. The doctor taps to confirm or fix each line. Nothing is spoken to the patient until it's confirmed.
5. **Read back and teach-back.** The confirmed prescription is read to the patient in their language, using plain words ("one tablet in the morning after food, one at night, for 5 days"). Then the agent asks the patient to say it back in their own words, translates that for the doctor, and flags any mismatch.
6. **Take it home.** The patient gets a voice note of the prescription in their language (WhatsApp or SMS link, or a QR printed on the slip) plus a picture card (sun / moon icons, tablet counts) that doesn't need reading.

Features: what we build this weekend vs later

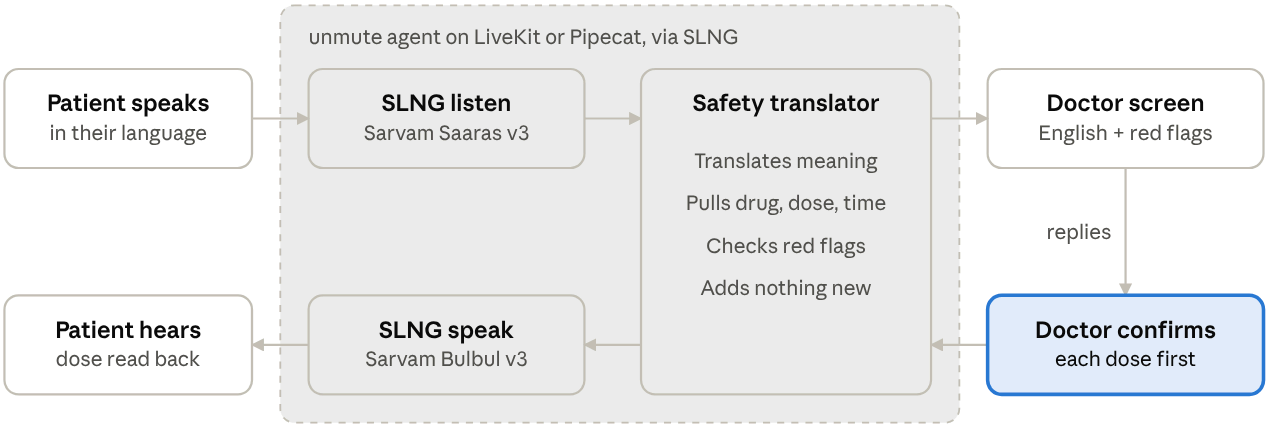
Feature	Hackathon MVP	Later
Two-way live interpretation, Kannada / Tamil / Hindi ↔ English	Yes	All 22 scheduled languages
Push-to-talk per side, language auto-detect + spoken confirmation	Yes	Hands-free with speaker separation
Prescription card: extract → doctor confirms → read back	Yes	Pull from hospital e-prescription / ABDM records
Teach-back check with mismatch flag	Yes	Track comprehension scores per clinic
Red-flag symptom alert	Basic keyword + LLM check	Clinically reviewed list per department
Take-home voice note + picture card	Voice note link	Printed QR on OPD slip, reminders by phone call
Consent and cost explanations	Fixed script	Hospital-specific consent forms

Feature	Hackathon MVP	Later
Pharmacy / nurse counter mode	–	Yes
Offline fallback phrase cards	–	Yes

How it's built

An unmute voice agent on SLNG: Indian-language listening and speaking through Sarvam models, a strict safety-translator step in the middle, and a doctor confirmation gate before anything clinical is spoken.

Every dose passes the doctor before the patient hears it



VaidyaVaani architecture · patient to doctor on top, doctor to patient below

The patient's words travel along the top row to the doctor; the reply travels back along the bottom. Saaras v3 can also translate speech straight to English, which lets the patient-to-doctor direction skip an LLM call and cut latency.

Layer	What we use	Why
Orchestration	unmute: the agent is written in YAML and compiled to LiveKit or Pipecat	Bonus points; one task per flow (consent, interpret, prescribe, teach-back); `unmute dev` shows streaming transcripts and per-request latency

Layer	What we use	Why
Listen, Indian languages	`sarvam/saaras:v3` via SLNG: 22 Indian languages plus English, with code-mix and translate-to-English modes (Sarvam docs)	Handles Kanglish and Tanglish; built for voice agents
Listen, Hindi / English	`slng/deepgram/nova:3-hi` and `nova:3-en`, SLNG-hosted (SLNG models)	Low latency, and a second provider to fail over to
Think	Any LLM, with a strict prompt: translate meaning, never add facts, return medicines as JSON fields	Keeps drug names and numbers structured and checkable
Speak	`sarvam/bulbul:v3` via SLNG: 11 Indian languages, 30+ voices	Natural voices in the patient's language
Failover	SLNG LiveKit plugin `connections=[...]` across models (PyPI)	Call keeps going if a provider drops
Front end	One web page: two talk buttons, live transcript, prescription card, QR for the voice note	Works on any phone or tablet

One thing to check with the SLNG team on the day: which deployment region keeps the whole pipeline in India, since that's our privacy promise.

Edge cases and how we handle them

The rule behind every answer below: when unsure, the agent asks or flags. It never guesses on anything that changes treatment. "MVP" marks what we handle this weekend.

1. Language and speech

Edge case	Example	How we handle it	MVP
Code-mixed speech	"Nanage two days inda fever ide, BP tablet already tagotini"	Saaras v3 has a code-mix mode built for this; English medical words are kept as spoken	Yes

Edge case	Example	How we handle it	MVP
Wrong language detected	Telugu speaker detected as Kannada	Agent confirms the language aloud at the start; one-tap language switch on screen	Yes
Language or dialect not supported	Tulu, Kodava, Bhojpuri, Lambani (not among the 22 scheduled languages)	Agent says so honestly and offers the patient's second language (most speak some Kannada, Hindi or Tamil); logs the request so the hospital sees demand	Yes
Regional dialect	North Karnataka Kannada, Dharwad Kannada, Chennai vs Madurai Tamil	Saaras claims dialect and accent coverage; we test with team members' dialects and log low-confidence turns	Partly
Dosing shorthand and local units	"1-0-1", "one spoon", "half tablet", "before food"	Converted to plain-language structured fields (morning / afternoon / night, before / after food) and read back explicitly	Yes
Lay terms for conditions	"sugar" (diabetes), "BP", "gas", "body heat", "weakness"	Glossary prompt: translate meaning, show the lay term in quotes to the doctor so they can ask follow-ups	Yes
Numbers that sound alike	15 vs 50 mg, "fifteen" vs "fifty"	Doses are never auto-spoken; shown as digits for the doctor to confirm first	Yes
Elderly, soft or slurred speech; speech impairment	Stroke patient	Ask to repeat once; then fall back to yes/no questions and on-screen picture choices	Later

Edge case	Example	How we handle it	MVP
Deaf or hard-of-hearing patient	–	Large-text display of the translation in the patient's script	Later

2. Audio and environment

Edge case	Example	How we handle it	MVP
Noisy OPD	Crowd, fans, crying babies	Push-to-talk instead of always-on listening; phone held close to the speaker	Yes
Both talk at once / interruptions	Doctor cuts in	Only the pressed side is recorded; a new press stops the agent speaking (barge-in)	Yes
Relative speaks instead of patient	Son answers for his mother	Fine: whoever holds the patient button is "patient side"; the doctor sees it's translated speech either way	Yes
Doctor talks to the nurse, not the patient	"Get the BP cuff"	Doctor holds the button only for things meant for the patient	Yes
Weak or no internet	Basement OPD	Clear "offline" state; fall back to pre-recorded phrase cards for the 30 most common instructions	Later

3. Medical safety (the part judges will probe)

Edge case	Example	How we handle it	MVP
Model invents or alters content	A drug or symptom that was never said	Translator prompt forbids adding information; doctor sees original-language transcript next to the English; prescription comes only from doctor-confirmed fields	Yes
Negation flipped	"No chest pain" → "chest pain"	Negations kept as a separate field in the extraction; teach-back catches the rest	Yes
Sound-alike drug names	Two similar brand names	Drug names shown as text for the doctor to confirm; the read-back to the patient uses the confirmed spelling, never the raw transcript	Yes
Emergency mid-consultation	Patient mentions crushing chest pain while describing "gas"	Red-flag alert on the doctor's screen immediately, in English	Yes
Patient asks the agent for medical advice	"Should I stop my BP tablet?"	Agent never answers medically; it translates the question to the doctor	Yes
Patient didn't understand	Teach-back answer doesn't match	Mismatch flagged to the doctor; agent repeats in simpler words	Yes
Sensitive topics with family in the room	Sexual health, pregnancy, mental health	Doctor can ask relatives to step out because the tool removes the need for them to interpret	Yes (by design)

Edge case	Example	How we handle it	MVP
Self-harm or abuse disclosed	–	Flagged to the doctor as urgent; the agent doesn't counsel, it translates	Yes
Child patient	Mother speaks for child	Same flow; dosing by weight stays the doctor's call, confirmed on screen	Yes

4. Privacy, consent and law

Edge case	How we handle it	MVP
Consent	Asked in the patient's language at the start; patient can decline and the doctor proceeds without it	Yes
Recording and storage	No audio stored by default; transcript kept only if consented, auto-deleted after the visit in the demo	Yes
Data location	Indian-language models via Sarvam run in India per SLNG's model list; SLNG's pitch is region locks and data residency. We confirm the India routing with the SLNG team on the day	Yes
India's data law	The DPDP Rules were notified on 13 November 2025 and phase in through 2027; we design for consent, purpose limits and deletion from day one (Hogan Lovells)	Yes (design)
Liability	Positioned as an assistive interpreter. The doctor confirms every clinical instruction; the agent never diagnoses or prescribes	Yes

5. Technical

Edge case	How we handle it	MVP
Latency too high for a busy OPD	Streaming STT and TTS through SLNG; LLM step kept short; we measure turn latency per request	Yes

Edge case	How we handle it	MVP
A speech provider goes down	SLNG failover across models (`connections=[...]` in the LiveKit plugin); demo this live	Yes
Wrong voice gender or age for the patient	Pick a neutral, slower Bulbul voice; let staff change it	Yes
Credits run out mid-demo	Watch usage in the SLNG dashboard; recorded backup demo ready	Yes
Cost at scale	Per-minute pricing; we show cost per consultation	Yes

6. People and adoption

Edge case	How we handle it	MVP
Patient distrusts a machine	The doctor stays in charge and visible; the voice speaks the patient's own language warmly and slowly	Yes
Doctor has no time	Push-to-talk adds seconds, not minutes; it replaces the hunt for a translator	Yes
Language politics	Product is language-neutral: Kannada is fully supported and is not forced on anyone	Yes
Staff never trained	One-screen UI with two buttons and pictures; no login for the patient	Yes

Metrics to show the judges

The brief asks for real numbers on latency, cost and audio quality before and after; measure these on the day, don't estimate them.

Metric	How to measure	Target to aim for
Turn latency (patient stops → doctor hears translation)	`unmute dev` shows per-request latency for every turn; compare provider called directly vs through SLNG	Fast enough for natural back-and-forth; report the median and worst case
Translation accuracy	20 recorded symptom sentences per language (Kannada, Tamil, Hindi), scored by a native speaker on the team: meaning kept yes/no	Report % kept
Critical errors	Same test set plus 10 prescriptions: any wrong drug, dose, timing or flipped negation	Zero reach the patient (all caught at doctor confirmation)
Teach-back catch rate	Seed 5 deliberately misunderstood read-backs; count how many get flagged	All 5
Cost per consultation	SLNG per-minute pricing × a 10-minute consult, plus LLM tokens	Show in rupees next to the cost of a human interpreter
Failover	Kill one STT provider mid-call	Call continues; report the gap in seconds
Steps automated	Count of manual steps removed (find a translator, write instructions, re-explain)	Before vs after flow on one slide

Demo script (2 minutes) and judge Q&A

Open with one patient's story, run one live consultation, close on the numbers.

Demo script

1. **0:00–0:20, the hook.** "In a Bengaluru study published this July, half the patients believed they'd been misdiagnosed because of language. One woman took her tablets at the wrong time for days until her daughter read the slip."
2. **0:20–1:20, live consultation.** Teammate A is a Tamil-speaking patient with fever and burning urination. Teammate B is the doctor, speaking English. Show: language confirm, two exchanges, a red-flag moment ("also some back pain and shivering"), the doctor dictating a prescription, the confirmation card, the read-back in Tamil, and the teach-back where the patient repeats it.
3. **1:20–1:40, the safety moment.** The doctor says "fifty" when the card reads 15; the card catches it before anything is spoken.
4. **1:40–2:00, under the hood.** One slide: unmute agent, SLNG models per language, latency and cost numbers, failover demo.

Keep a recorded version of step 2 ready in case the Wi-Fi fails.

Questions judges will ask

Question	Answer
Why not Google Translate?	It's text-first, one direction, no dose safety, and staff in the study stopped using it because patients couldn't follow it. Ours is voice-first for low-literacy patients and confirms every dose.
What if the AI gets it wrong?	Nothing clinical is spoken until the doctor confirms it, and teach-back checks the patient understood. The agent never diagnoses or advises.
Isn't this a liability risk?	Less than the status quo: today a security guard interprets. We're an assistive tool with the doctor in control and a visible transcript.
What about privacy?	Consent first, no audio stored by default, Indian-language models running in India, designed for the DPDP rules.
Who pays?	Hospitals and state health departments per consultation minute; later, integration into e-prescription and Ayushman Bharat digital records.

Question	Answer
Languages you don't support?	We say so honestly, offer the patient's second language, and log demand.
Why SLNG?	One API across Sarvam (Indian languages) and other providers, failover, region control, and unmute let us build the agent in YAML in a weekend.

Sources

- Lost in translation: language barriers in Karnataka healthcare (Frontiers in Communication, July 2026)
- 107 languages spoken in Bengaluru (Udayavani)
- KLDA proposal on Kannada prescriptions (South First)
- Doctor confronted over Kannada at Minto Hospital (The News Minute)
- Protest disrupts Victoria Hospital OPD (Deccan Herald)
- Google Translate for discharge instructions (Becker's, on the JAMA study)
- AI transcription inventing words in hospitals (AP via ABC News)
- Bhashini and NHA partnership (Digital Health News)
- DPDP Act and Rules in force (Hogan Lovells)
- SLNG supported models
- SLNG plugin for LiveKit (PyPI)
- unmute releases (GitHub)
- Sarvam Saaras v3 model docs
- How SLNG uses Sarvam for Indic voice (Sarvam)